

The ν PGCM: A Flexible Finite Element Model of the Large-Scale Ocean Circulation

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Abstract**Plain Language Summary**

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1 Introduction

The ocean is a vast reservoir of heat, carbon, nutrients, and other tracers crucial to the evolution of Earth's climate (e.g. Sarmiento & Gruber, 2006). The dense abyssal waters formed due to buoyancy loss in the polar regions (e.g., Lumpkin & Speer, 2007; Talley, 2013) sequester these tracers away from the surface for hundreds to thousands of years, regulating the climate on long timescales (e.g., Sarmiento & Toggweiler, 1984). Alongside observational efforts, numerical ocean models are valuable tools for elucidating the key elements of this circulation. Historically, improvements in these models have gone hand-in-hand with developments in our oceanographical understanding and have shaped policy and planning (Fox-Kemper et al., 2021). In particular, the character of the modeled overturning circulation exerts a strong control on simulated past (e.g., Kageyama et al., 2009; Zhang et al., 2013), present, and future (e.g., Gregory et al., 2005; Winton et al., 2014; He et al., 2017; Weijer et al., 2020; Baker et al., 2023) climates.

Paradoxically, this slow, planetary-scale circulation is dependent on fast, small-scale turbulence that cannot be directly resolved in a numerical model of the global ocean. The densest bottom waters formed in the Southern Ocean must undergo diapycnal mixing to lighten and return to the surface (e.g., W. H. Munk, 1966; W. Munk & Wunsch, 1998; Ferrari, 2014). Typically, this small-scale mixing is parameterized in models by down-gradient diffusion of buoyancy with either a fixed profile of turbulent diffusivity κ or a time-dependent scheme calibrated to high-resolution simulations (e.g., Large et al., 1994). Microstructure observations of turbulence in the abyss show that mixing is strongly heterogeneous and most vigorous over rough topography (e.g. Polzin et al., 1997; Ledwell et al., 2000; Waterhouse et al., 2014), where internal waves break and become non-linear (e.g., Garrett & Kunze, 2007; Nikurashin & Ferrari, 2011). This spatial structure of turbulence suggests that the abyssal upwelling crucial for closing the overturning is confined to the bottom $O(10\text{ m})$ of the ocean and over topographic slopes (e.g., Ferrari et al., 2016; de Lavergne et al., 2016; McDougall & Ferrari, 2017).

These thin bottom boundary layers (BLs) present a unique challenge in simulating the large-scale ocean circulation. Due to computational demands, even state-of-the-art global ocean models

have a resolution of $O(100\text{ m})$ near the bottom (e.g., ECCO-Consortium et al., 2021), with yet coarser
 40 resolutions needed to simulate the coupled climate system over thousands of years (e.g., Rugenstein
 et al., 2019). When simulating the full Boussinesq system, the timestep is limited by fast-timescale
 dynamics such as small-scale turbulence, internal waves, and baroclinic eddies, most of which can-
 not be directly resolved at the coarse spatial resolution of global ocean models in the first place.
 A more natural approach would be to employ the planetary geostrophic (PG) approximation (e.g.,
 45 Pedlosky, 1979; Vallis, 2017), filtering out the inertial terms in the momentum equations by as-
 suming small Rossby numbers while retaining full buoyancy advection by assuming large horizontal
 scales. The PG equations have been a cornerstone of our understanding of the large-scale ocean cir-
 culation (e.g., Robinson & Stommel, 1959; Welander, 1959; Colin de Verdière, 1988; Samelson &
 Vallis, 1997a; Salmon, 1998; Pedlosky, 1998), with a number of PG circulation models (PGCMs)
 50 already in existence (e.g., Salmon, 1986; Samelson & Vallis, 1997b; Edwards et al., 1998; Callies &
 Ferrari, 2018). For computational simplicity, these models employ Rayleigh drag in the momentum
 equations, leading to excessive damping of interior flows and unphysical bottom BLs (Peterson &
 Callies, 2023). In our ν PGCM, described below, we instead view the PG flow as the residual flow af-
 ter a thickness-weighted average over transients, parameterizing their effects on momentum through
 55 an Eliassen–Palm flux with turbulent viscosity ν (e.g., Young, 2012; Jansen et al., 2024).

The complex land-sea boundary of the ocean poses another challenge for its numerical sim-
 ulation. Not only is this geometry crucial in setting up the qualitative “figure-8” shape of the over-
 turning (e.g., Ferrari et al., 2014), but in the context of abyssal upwelling, the sloping bottom is
 what allows bottom BL flows to upwell dense water against the stratified buoyancy field. Idealized
 60 “shoebox” models with a flat bottom are capable of generating seemingly Earth-like overturning cir-
 culations (e.g., Nikurashin & Vallis, 2011; Jansen & Nadeau, 2019), but the diapycnal upwelling of
 abyssal waters occurs within unrealistic BLs on vertical side-walls in these models. The majority
 of ocean models with realistic bathymetry use structured meshes and finite differencing (Arakawa
 & Lamb, 1981) due to their ease, efficiency, and familiarity (e.g., Griffies et al., 2009; Fox-Kemper
 65 et al., 2019). Although shaved-cell implementations exist to better resolve the bottom bathymetry
 (e.g., Adcroft et al., 1997), they are rarely used in practice and still lead to a “staircase” pattern that
 requires prohibitively high horizontal resolution to represent up-slope BL flows. Terrain-following
 vertical coordinates are an attractive solution, though errors associated with representing horizontal
 pressure gradients in these coordinates (e.g., Haney, 1991) and the singularity along the coastline are
 70 currently a barrier for their adoption in global models.

While less standard in models of the global ocean, the geometrical flexibility of unstructured grids and the high order accuracy of the finite element method combine to make a particularly desirable alternative. Part of the reason for the slow adoption of these methods in geophysical fluid dynamics problems is that they were initially designed for elliptic partial differential equations. The PG equations can be thought of separately as an advection–diffusion equation for buoyancy and an inversion statement for the flow and pressure. Finite element solutions to the advection–diffusion problem are now standard, though a stabilization scheme is sometimes necessary (e.g., Elman et al., 2014). The PG inversion, on the other hand, can be thought of as an incompressible Stokes problem with a body force due to rotation and the added physics of buoyancy forces. Textbook mixed finite element techniques (e.g., Hughes, 1987; Elman et al., 2014) are effective at solving the Stokes problem for isotropic diffusion, but they lose accuracy and even destabilize for hydrostatic problems (e.g., Guillén-González & Rodríguez-Galván, 2015). Following multiple other studies (e.g., Kuang et al., 2005; Garner et al., 2007; Salmon, 2009), we employ the “aspect ratio trick,” artificially increasing the aspect ratio α while holding all other nondimensional parameters fixed. This re-introduces diffusion in the local-vertical momentum equation, guaranteeing numerical stability. Even for large aspect ratios ($\alpha \lesssim 1/2$) compared to the ocean ($\alpha = 10^{-3}$), the qualitative behavior of the flow remains the same.

In the next section, we introduce the PG equations that we use to describe the physics of the mixing-generated abyssal circulation. We will choose a particular nondimensionalization that makes the role of the aspect ratio explicit while also retaining isotropy in the friction terms. In section 3, we describe how the ν PGCM solves these equations numerically using finite elements and an artificially enhanced aspect ratio. We briefly conclude with some example simulations in a parabolic bowl basin in section 4, showcasing some of the capabilities of this model.

2 Planetary geostrophic equations

As prefaced in the introduction, we aim to describe the large-scale ocean circulation using the PG equations, which are derived from the Boussinesq equations by assuming large horizontal scales and small Rossby numbers. In index notation, the dimensional PG equations in Cartesian space (x_1 ,

x_2, x_3) read

$$2e_{ijk}\Omega_j u_k = -\frac{\partial p}{\partial x_i} + bz_i + \frac{\partial}{\partial x_j} (2\nu\sigma_{ij}), \quad (1)$$

$$\frac{\partial u_i}{\partial x_i} = 0, \quad (2)$$

$$\frac{\partial b}{\partial t} + u_i \frac{\partial b}{\partial x_i} = \frac{\partial}{\partial x_i} \left(\kappa \frac{\partial b}{\partial x_i} \right), \quad (3)$$

where $\mathbf{u} = (u_1, u_2, u_3)$ is the velocity vector, p is the pressure, and b is the buoyancy. The first term in (1) represents the Coriolis acceleration, with e_{ijk} the Levi-Civita symbol for the cross product and Ω the rotation vector of the volume; we do not make the traditional approximation. The buoyancy force acts only in the direction opposite to gravity, defined by the unit vector $\mathbf{z}(\mathbf{x})$. We parameterize turbulent mixing of buoyancy by a down-gradient flux proportional to the turbulent diffusivity κ (W. H. Munk, 1966). The Eliassen-Palm fluxes parameterizing mixing due to eddies contribute a diffusion term in the momentum equations with eddy viscosity ν . To account for spatially varying ν , this friction term is written in terms of the rank-2 strain rate tensor,

$$\sigma_{ij} = \sigma(\mathbf{u})_{ij} = \frac{1}{2} \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right). \quad (4)$$

For constant ν , this term is equivalent to the regular Laplacian of $\mathbf{u}$. We apply a no-slip condition on the flow at the bottom boundary of the domain Γ_B , and at the surface Γ_S we demand no normal flow and that the stress in the local horizontal direction is set by the wind stress:

$$\mathbf{u} = 0 \quad \text{on} \quad \Gamma_B, \quad (5)$$

$$u_i n_i = 0 \quad \text{and} \quad \nu \sigma_{ij} n_i = \tau_{ij} n_i \quad \text{on} \quad \Gamma_S, \quad (6)$$

where $\mathbf{n}$ is the normal vector to the boundary. The buoyancy flux through the bottom is set to $\mathcal{G}$ (typically zero unless accounting for geothermal heating effects), and either the buoyancy or the buoyancy flux can be set at the surface:

$$-\kappa \frac{\partial b}{\partial x_i} n_i = \mathcal{G} \quad \text{on} \quad \Gamma_B, \quad (7)$$

$$b = b_S \quad \text{or} \quad -\kappa \frac{\partial b}{\partial x_i} n_i = \mathcal{F} \quad \text{on} \quad \Gamma_S. \quad (8)$$

95 These PG dynamics can be viewed separately as an evolution equation for buoyancy (3) and
 an inversion statement for the flow (1). A single timestep in the numerical model, described below
 and illustrated in Fig. 4, makes use of this separation rather than simultaneously evolving the entire
 system. Although the code supports arbitrary initial conditions, for the purposes of understanding
 the basic phenomenology of the mixing-driven circulation, we here initialize all simulations with
 100 flat isopycnals aligned with gravity and constant stratification N^2 such that $\partial_z b = N^2$ at $t = 0$ (using
 the notation $\partial_z \equiv z_i \partial_{x_i}$).

2.1 Nondimensionalization and parameters

To isolate role of the aspect ratio in this problem, the vPGCM ultimately solves the nondimensional PG equations, derived below. We scale all spatial coordinates by the natural length scale of the domain L (e.g., the radius of the planet or the width of the basin) and all velocities by the same scale:

$$x_i = L\tilde{x}_i \quad \text{and} \quad u_i = U\tilde{u}_i, \quad (9)$$

for $i = 1, 2, 3$. Given the typical initial condition defined above, a natural scaling for buoyancy would be $b \sim N^2 H$ for some depth scale of the ocean H . Unlike in quasi-geostrophic theory, however, the PG equations do not explicitly impose a background stratification so that, in general, a representative scale for N^2 in the abyssal ocean will depend on the context of the problem. We additionally define characteristic scales for the rotation rate and mixing coefficients:

$$\Omega_i \sim \Omega, \quad \nu \sim \nu_0, \quad \kappa \sim \kappa_0. \quad (10)$$

Finally, we assume that the pressure gradient term in (1) scales with the Coriolis term, that the buoyancy also scales with the pressure scale divided by H from hydrostatic balance, and that time scales advectively:

$$p \sim \Omega U L, \quad b \sim \frac{\Omega U L}{H} = N^2 H, \quad t \sim \frac{L}{U}. \quad (11)$$

Applying these scales to equations (1)–(3) yields the following nondimensional PG equations:

$$2e_{ijk}\tilde{\Omega}_j\tilde{u}_k = -\frac{\partial\tilde{p}}{\partial\tilde{x}_i} + \alpha^{-1}\tilde{b}z_i + \alpha^2\epsilon^2\frac{\partial}{\partial\tilde{x}_j}(2\tilde{\nu}\tilde{\sigma}_{ij}), \quad (12)$$

$$\frac{\partial\tilde{u}_i}{\partial\tilde{x}_i} = 0, \quad (13)$$

$$\mu\varrho\left(\frac{\partial\tilde{b}}{\partial\tilde{t}} + \tilde{u}_i\frac{\partial\tilde{b}}{\partial\tilde{x}_i}\right) = \alpha^2\epsilon^2\frac{\partial}{\partial\tilde{x}_i}\left(\tilde{\kappa}\frac{\partial\tilde{b}}{\partial\tilde{x}_i}\right), \quad (14)$$

where $\alpha = H/L$ is the aspect ratio, $\epsilon^2 = \nu_0/\Omega H^2$ is the Ekman number, $\varrho = N^2 H^2/\Omega^2 L^2$ is the Burger number, and $\mu = \nu_0/\kappa_0$ is the turbulent Prandtl number. The wind stress boundary condition at the surface is now $\alpha^2\epsilon^2\tilde{\nu}\tilde{\sigma}(\tilde{u})_{ij}n_i^S = \tilde{\tau}_{ij}n_i^S$ and the buoyancy boundary conditions are $-\tilde{\kappa}\partial_{\tilde{x}_i}\tilde{b}n_i = \tilde{G}$ at the bottom and either $\tilde{b} = \tilde{b}_S$ or $-\tilde{\kappa}\partial_{\tilde{x}_i}\tilde{b}n_i = \tilde{F}$ at the surface. Since we will work in nondimensional coordinates for the remainder of the paper, we will henceforth drop the $\sim$ notation.

With all three spatial coordinates scaled by L , the effect of the aspect ratio α on the dynamics is made explicit and the domain $\mathcal{D}$ itself must have an aspect ratio of α . For an isolated basin with x_3 being the vertical coordinate aligned with gravity, this implies that $-\alpha \leq x_3 \leq 0$ (Fig. 1a). If instead $\mathcal{D}$ is the entire ocean, α naturally becomes the thickness of the shell relative to the radius of the sphere (Fig. 1b). This scaling guarantees that the viscous friction term in (12) is spatially isotropic,

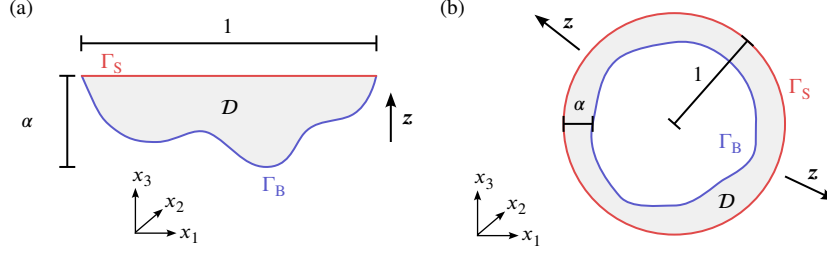


Figure 1. Sketches of a slice along x_1 of (a) an isolated basin with x_3 aligned with gravity and (b) the global ocean represented as a spherical shell embedded in Cartesian space.

a desirable property for computing numerical solutions. More importantly, for $\alpha > 0$ hydrostatic balance is not exactly required, as can be seen by dotting (12) with the local vertical z :

$$\frac{\partial p}{\partial z} = \alpha^{-1} b + \alpha^2 \epsilon^2 \frac{\partial}{\partial x_j} (2\nu \sigma_{ij}) z_i, \quad (15)$$

where again $\partial_z \equiv z_i \partial_{x_i}$. For the ocean, typical order-of-magnitude length scales are $H \approx 10^3$ m and $L \approx 10^6$ m, implying $\alpha \approx 10^{-3}$. Hence, the small aspect ratio assumption is often made, eliminating the diffusion term in (15). While standard finite element techniques may be used to solve the Stokes problem with rotation, they become brittle under this approximation (e.g., Guillén-González & Rodríguez-Galván, 2015). To leverage established methods, we will therefore keep α larger than zero but small enough to capture the qualitative dynamics of the ocean. A similar approach was taken by Salmon (1986) for a PG model with Rayleigh drag. In section 3, we will explore the effects of an enhanced aspect ratio on the dynamics of our Fickian diffusion model.

The other parameters of the flow are free to be chosen to match the physical context. The rotation rate of the Earth is $\Omega \approx 10^{-4} \text{ s}^{-1}$ and the stratification in the abyss is around $N \approx 10^{-3} \text{ s}^{-1}$, yielding a Burger number of $\rho \approx 10^{-4}$. Over rough topography, one might expect strong turbulence associated with a turbulent diffusivity on the order of $\kappa_0 \approx 10^{-3} \text{ m}^2 \text{ s}^{-1}$ (e.g., Waterhouse et al., 2014). The magnitude of the turbulent viscosity depends on whether or not a parameterization of eddies is considered. Without eddies, it is reasonable to assume that, for weakly stratified abyssal waters, small-scale mixing of buoyancy would occur on similar scales to the mixing of momentum, implying that $\nu_0 \sim \kappa_0$, or $\mu \sim 1$ (e.g., Caulfield, 2021). An Eliassen–Palm flux equivalent to an eddy diffusivity of $K \approx 10^3 \text{ m}^2 \text{ s}^{-1}$ (Gent & McWilliams, 1990), however, would require an enhanced viscosity of $\nu_0 \approx 10 \text{ m}^2 \text{ s}^{-1}$, or $\mu \approx 10^4$. This is consistent with previous studies that find $\mu \gg 1$ in the presence of submesoscale baroclinic eddies generated in abyssal mixing layers (e.g., Wenegrat et al., 2018; Callies, 2018). In the first case, where $\nu_0 \approx 10^{-3} \text{ m}^2 \text{ s}^{-1}$, the Ekman number ϵ is on the order

of 10^{-3} . In the eddy-parameterizing case of $\nu_0 \approx 10 \text{ m}^2 \text{ s}^{-1}$, on the other hand, it is enhanced to $\varepsilon \approx 10^{-1}$. Given that the nondimensional bottom Ekman layer thickness $\delta/L = \sqrt{2\nu_0/(\Omega_0 L^2)} = \sqrt{2}\alpha\varepsilon$, these differences are crucial in setting the minimum resolution required for numerical simulations. In the following, we apply the eddy parameterization with $\varepsilon = 2 \times 10^{-2}$ and $\mu = 10^4$ so that $\mu\rho = 1$, which is both more realistic and reduces resolution constraints.

3 Numerical method

In this section, we describe the numerical scheme used in the vPGCM to solve the nondimensional PG equations (12)–(14). We discretize the domain using an unstructured mesh of tetrahedra, allowing considerable geometrical flexibility at the cost of requiring a more sophisticated numerical method than standard finite differences on a structured grid. The numerical solution satisfies the weak Galerkin form of the equations and lives in a specially chosen finite element space to guarantee stability. To obtain smooth solutions, we must artificially increase the aspect ratio α , introducing diffusion in the hydrostatic equation. The matrix equations for the PG inversion and evolution are solved separately using iterative solvers. We use Strang splitting to handle advection and diffusion as separate partial steps in the evolution equation.

3.1 Weak formulation

To derive the finite element formulation of the model equations, we first define the function spaces in which we would like our solutions to live. These spaces need not specify any Neumann conditions such as in (6), (7), and (8), as those will be taken care of by boundary integrals in the weak formulation, as we will see below. The velocity, pressure, and buoyancy spaces are then:

$$\mathcal{U} \equiv \left\{ \mathbf{u} \in [H^1(\mathcal{D})]^3 : \mathbf{u} = 0 \text{ on } \Gamma_B \text{ and } u_i n_i = 0 \text{ on } \Gamma_S \right\}, \quad (16)$$

$$\mathcal{P} \equiv \left\{ p \in L^2(\mathcal{D}) : \int_{\mathcal{D}} p \, d\mathbf{x} = 0 \right\}, \quad (17)$$

$$\mathcal{B} \equiv L^2(\mathcal{D}) \quad \text{or} \quad \mathcal{B} \equiv \left\{ b \in L^2(\mathcal{D}) : b = b_0 \text{ on } \Gamma_S \right\}, \quad (18)$$

respectively, where the buoyancy space depends on whether the Dirichlet or Neumann condition is chosen in (8). Here, $[H^1(\mathcal{D})]^3$ refers to the Sobolev space of all three-component vector functions that satisfy

$$\|\mathbf{u}\|_{H^1}^2 = \int_{\mathcal{D}} \left(u_i u_i + \frac{\partial u_i}{\partial x_j} \frac{\partial u_i}{\partial x_j} \right) d\mathbf{x} < \infty, \quad (19)$$

and $L^2(\mathcal{D})$ refers to the Lebesgue space of all functions that are square integrable, i.e.,

$$\|p\|_{L^2}^2 = \int_{\mathcal{D}} p^2 \, d\mathbf{x} < \infty. \quad (20)$$

Without the integral constraint in $\mathcal{P}$, the pressure may only be determined up to an additive constant.

The weak form of the PG inversion is obtained by dotting equations (12) and (13) with test functions $(v, q) \in \mathcal{U} \times \mathcal{P}$, integrating over the domain, and performing integration by parts on the divergence of the strain rate, yielding

$$\int_D \left[2e_{ijk}\Omega_j u_k v_i + \frac{\partial p}{\partial x_i} v_i + \frac{\partial u_i}{\partial x_i} q + \alpha^2 \epsilon^2 v \sigma(u)_{ij} \sigma(v)_{ij} \right] dx = \int_D \alpha^{-1} b z_i v_i dx + \int_{\Gamma_S} \tau_{ij} n_j v_i dx. \quad (21)$$

This is known as a saddle-point problem (notice that the pressure p never multiplies its test function q). As we will see in the next section, this limits the discrete spaces on which we may stably represent the solution. Following the same steps for the buoyancy equation (14), this time multiplying by a test function $c \in \mathcal{B}$, yields the weak formulation

$$\int_D \left[\frac{\partial b}{\partial t} c + u_i \frac{\partial b}{\partial x_i} c + \theta \kappa \frac{\partial b}{\partial x_i} \frac{\partial c}{\partial x_i} \right] dx = \theta \int_{\Gamma_B} \mathcal{G} c dx + \theta \int_{\Gamma_S} \mathcal{F} c dx, \quad (22)$$

145 where $\theta = \alpha^2 \epsilon^2 / \mu \rho$. In the Dirichlet boundary condition case, the last integral over Γ_S is removed .

3.2 Finite element discretization

To solve equations (21) and (22) numerically, we tessellate the domain into a mesh $\mathcal{T}_h$ of finite pieces (or “elements”) with characteristic length scale h . In two dimensional space, each element $T_k \in \mathcal{T}_h$ is a triangle whereas in three dimensions we use tetrahedra (Fig. 2). In general, the quality of the mesh can have a significant impact on the accuracy of the solution. We use meshes generated by Gmsh (Geuzaine & Remacle, 2009), although there are many other meshing software packages available. We here show results for a parabolic bowl with

$$\Gamma_B = \left\{ x \in \mathbb{R}^3 : x_1^2 + x_2^2 \leq 1 \text{ and } x_3 = -\alpha(1 - x_1^2 - x_2^2) \right\}, \quad (23)$$

$$\Gamma_S = \left\{ x \in \mathbb{R}^3 : x_1^2 + x_2^2 \leq 1 \text{ and } x_3 = 0 \right\}, \quad (24)$$

though arbitrary geometries are possible with this method. In general, one now defines a subspace of the full solution space that can be spanned by a finite number of basis functions defined on the mesh, converting the continuous form of the weak formulation (16)–(22) into a discrete problem. A simple and common choice for this subspace is the set of continuous, piecewise-polynomial functions of degree n over the elements, $P_n(\mathcal{T}_h)$. Denoting each node in the mesh by x_i , one can create a set of basis functions for this space $\{\varphi_i\}$ that satisfy

$$\varphi_i(x_j) = \delta_{ij}, \quad (25)$$

where δ_{ij} is the Kronecker delta. For the linear space $P_1(\mathcal{T}_h)$, the element vertices supply enough nodes to span the space (orange crosses in Fig. 2a,c), but for higher-order spaces, more nodes are needed.

As discussed above, this formulation of the PG inversion is equivalent to the Stokes problem with rotation, allowing us to employ a standard mixed finite element scheme. Although not all discrete subspaces are stable for saddle-point problems such as (21), if they satisfy the so-called LBB condition, a unique solution that depends continuously on the forcing exists (e.g. Hughes, 1987; Elman et al., 2014). It is possible to choose a finite element basis that does not satisfy the LBB condition, but ad hoc stabilization schemes are necessary (e.g., Danilov et al., 2004). We instead choose the simple and accurate P_2 – P_1 basis that is known to satisfy the LBB condition. In this basis, the velocities are quadratic while the pressure is linear, so that the discrete subspaces defined over the mesh are

$$\mathcal{U}_h \equiv [P_2(\mathcal{T}_h)]^3 \cap \mathcal{U} \quad \text{and} \quad \mathcal{P}_h \equiv P_1(\mathcal{T}_h) \cap \mathcal{P}. \quad (26)$$

The degrees of freedom for the velocity components therefore exist on both the midpoints and vertices of the elements while those of the pressure are just on the vertices (Fig. 2a,c). The added degrees of freedom from using second-order elements increases the computational demand of the PG inversion, but, as we will see in the next section, the convergence rate is rapid enough that resolution constraints are not as high. While not needed for stability, we represent buoyancy with quadratic, rather than linear, polynomials to ensure high accuracy:

$$\mathcal{B}_h \equiv P_2(\mathcal{T}_h) \cap \mathcal{B}. \quad (27)$$

With the discrete subspaces defined in (26) and (27), we can assemble the matrices needed to compute the PG inversion and evolve buoyancy in time. If $\{(\varphi_i)_j\}$, $\{\psi_j\}$, and $\{\varphi_i\}$ are the sets of basis vectors for $\mathcal{U}_h$, $\mathcal{P}_h$, and $\mathcal{B}_h$, respectively, then the solution to the weak formulation of the PG equations can be represented as linear combinations of these functions:

$$(u_i)_h(\mathbf{x}) = (\mathbf{u}_i)_j(\varphi_i)_j(\mathbf{x}), \quad p_h(\mathbf{x}) = \mathbf{p}_j\psi_j(\mathbf{x}), \quad b_h(\mathbf{x}) = \mathbf{b}_j\varphi_j(\mathbf{x}), \quad (28)$$

with $\mathbf{u}_i$, $\mathbf{p}$, and $\mathbf{b}$ being the vectors of projection coefficients. Note that, because the underlying basis vectors are continuously defined over the entire domain, so are the solutions. This is in contrast to typical finite difference methods where the solution is only defined on the grid. Conveniently, because of the choice of nodal basis functions defined by the property (25), the projection coefficients are equal to the values of these functions on the nodes of the mesh:

$$(u_i)_h(\mathbf{x}_k) = (\mathbf{u}_i)_k, \quad p_h(\mathbf{x}_k) = \mathbf{p}_k, \quad b_h(\mathbf{x}_k) = \mathbf{b}_k. \quad (29)$$

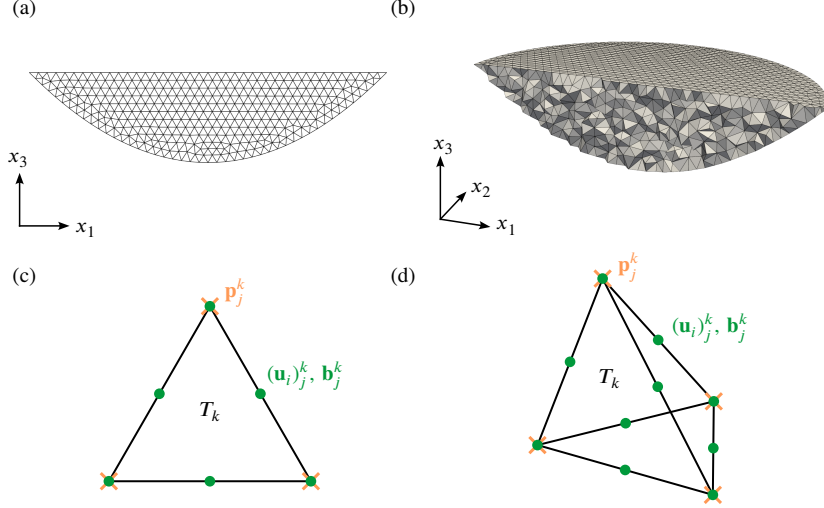


Figure 2. Meshes $\mathcal{T}_h$ of (a) two- and (b) three-dimensional basin domains and sketches of (c) triangular and (d) tetrahedral finite elements $T_k \in \mathcal{T}_h$. The meshes are generated for a parabolic basin with $\alpha = 1/2$ and a characteristic resolution of $h = 1/20$. In the mixed finite element method described in section 3.2, the pressure degrees of freedom $\mathbf{p}_j^k$ correspond to values on the vertices of each element while the velocity and buoyancy degrees of freedom $(\mathbf{u}_i)_j^k$ and $\mathbf{b}_j^k$ exist on both the vertices and midpoints. The superscripts indicate that these are the degrees of freedom on the local element T_k .

Given the buoyancy coefficients $\mathbf{b}$, the coefficients for the velocity and pressure $\mathbf{x} = [\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3, \mathbf{p}]^T$ can be determined by solving the matrix equation

$$\hat{\mathbf{K}}\mathbf{x} = \hat{\mathbf{M}}\mathbf{b} + \mathbf{s}, \quad (30)$$

where $\hat{\mathbf{K}}$, $\hat{\mathbf{M}}$, and $\mathbf{s}$ are computed by integrating the weak formulation of the inversion (21) for each basis function. For instance,

$$\hat{\mathbf{M}}_{ij} = \int_D \alpha^{-1} \varphi_j z_k (\varphi_k)_i \, d\mathbf{x} \quad \text{and} \quad \mathbf{s}_i = \int_D \tau_{kj} n_j (\varphi_k)_i \, d\mathbf{x}. \quad (31)$$

150 We use the Julia (Bezanson et al., 2017) package Gridap.jl to automate this step (Badia & Verdugo, 2020). A high-resolution three-dimensional inversion can easily contain millions of tetrahedra, making a direct solve of the matrix equation (30) impractical. Instead, we load $\hat{\mathbf{K}}$ onto a GPU using CUDA.jl (Besard et al., 2019) and solve the problem iteratively using Krylov.jl’s (Montoisson & Orban, 2023) implementation of the generalized minimum residual method (GMRES). In the following
155 section, evaluate the error of this method.

3.3 Convergence rates for the PG inversion

As a test case to validate the accuracy and convergence rate of the numerical inversion, we solve for $(\mathbf{u}_h, p_h)$ given flat isopycnals in the parabolic bowl geometry: $b = \alpha^{-1}x_3$. The exact solution is

$$\mathbf{u} = 0 \quad \text{and} \quad p = \frac{x_3^2}{2\alpha^2} - C, \quad (32)$$

where C is a constant to satisfy the integral constraint equal to $4/35$ for the two-dimensional bowl and $1/12$ for the three-dimensional bowl. Deviations from this exact solution result from inaccuracies in representing horizontal pressure gradients, making it a useful test case. Since isopycnals are flat throughout much of the ocean, these pressure gradient errors can be detrimental for ocean models if they are not small relative to full flow. Such errors are ubiquitous in terrain-following coordinate models (e.g., Haney, 1991), prohibiting their use for global-scale problems. As we will see below, these errors are relatively modest in our model and, importantly, their magnitude rapidly decreases with increasing resolution.

From finite element theory (e.g., Hughes, 1987; Elman et al., 2014), the error in the so-called “energy norm” for the P_2 – P_1 method scales as

$$\|\mathbf{u}_h\|_{H^1} + \|p - p_h\|_{L^2} \sim O(h^2), \quad (33)$$

where, again, h is the characteristic mesh resolution. This is indeed the case for a range of parameters in both two- and three-dimensional inversions (Fig. 3a,b). We also find that this error scales roughly like $O(\alpha^{-2}\epsilon^{-2})$. This follows from the fact that the BL scale is proportional to $\alpha\epsilon$ so that the effective resolution with respect to the BL $h_{\text{eff}} \sim h/(\alpha\epsilon)$. Since the error tends to be concentrated near the boundary for this problem, it is not a surprise that the energy norm scales like h_{eff}^2 .

While there is not a general theory for the scaling of the maximum error $\|\mathbf{u}_h\|_{L^\infty} \equiv \sup_D |\mathbf{u}_h|$, its value is more interpretable than that of the energy norm. We find that, for this particular test with flat isopycnals, the error empirically scales with h^3 (Fig. 3c,d). This rapid convergence rate in the maximum pressure gradient error is promising for the accuracy of large-scale ocean circulation simulated by this model. Next, we will couple the inversion to the evolution equation to simulate time-dependent flow.

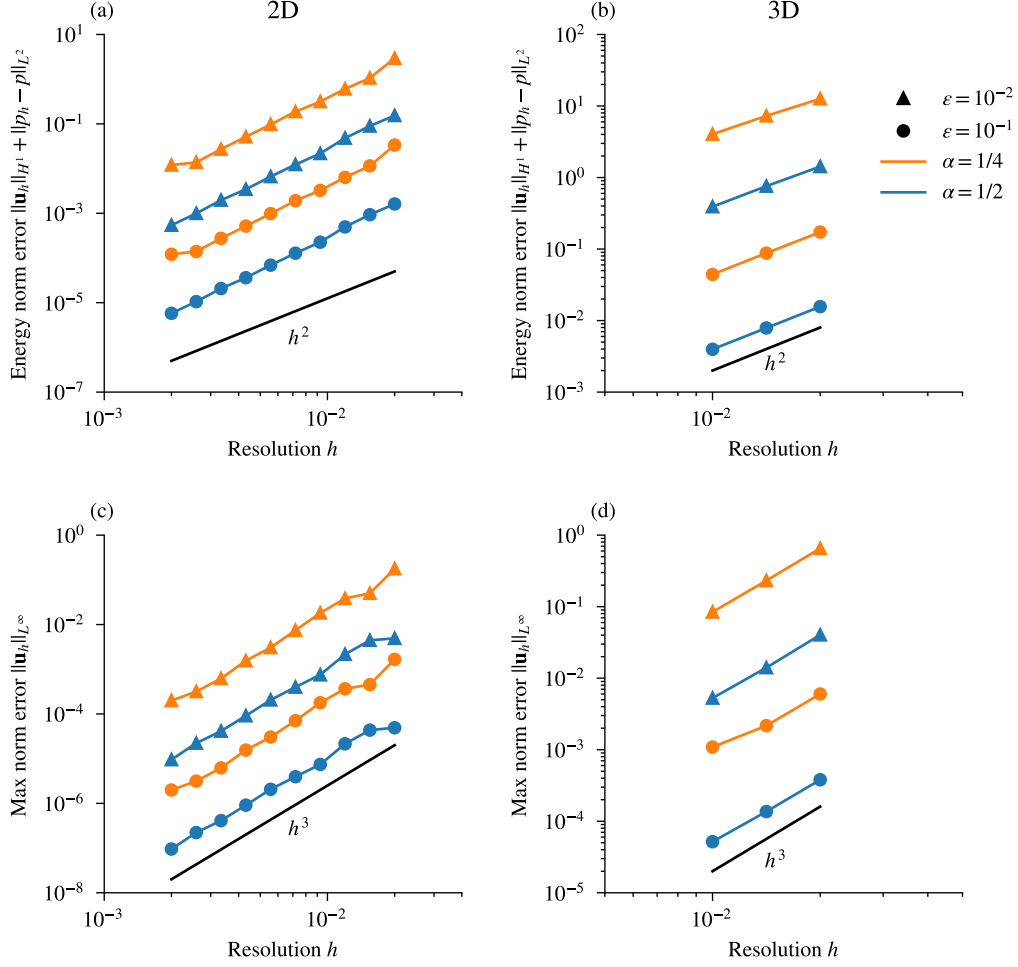


Figure 3. Convergence rates of the PG inversion with respect to resolution h , Ekman number ϵ , and aspect ratio α for the case of flat isopycnals in a parabolic bowl basin (see Fig. 2a,b) with true solution (32). (a) and (c) are for two-dimensional geometry and (b) and (d) show results for the three-dimensional inversion. The error scales like h^2 in the energy norm [(a) and (b)] and h^3 in the max norm [(c) and (d)].

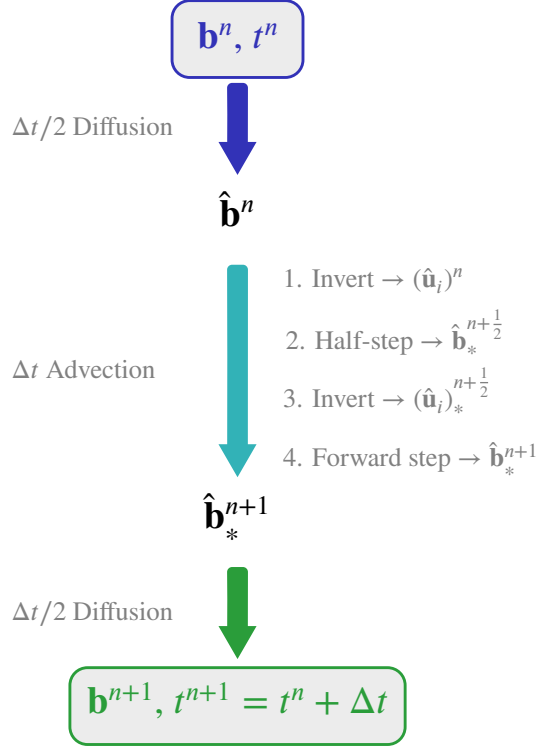


Figure 4. Flow chart of a single Strang-split timestep from t^n to $t^{n+1} = t^n + \Delta t$ in a simulation (see section 3.4 for details). “Invert” refers to solving the matrix equation (30) for the PG inversion. The half- and forward advection steps are described in equations (38) and (40), respectively, and the $\Delta t/2$ diffusion steps follow equation (41).

3.4 Timestepping

Once the solution is projected onto the discrete finite element space described above, the weak form of the buoyancy evolution (22) becomes

$$\mathbf{M} \frac{\partial \mathbf{b}}{\partial t} + F(\mathbf{u}_i, \mathbf{b}) + \theta \mathbf{K} \mathbf{b} = \theta \mathbf{f}, \quad (34)$$

where

$$\mathbf{M}_{ij} = \int_D \varphi_i \varphi_j \, d\mathbf{x} \quad \text{and} \quad \mathbf{K}_{ij} = \int_D \kappa \frac{\partial \varphi_i}{\partial x_k} \frac{\partial \varphi_j}{\partial x_k} \, d\mathbf{x}, \quad (35)$$

are known as the “mass” and “stiffness” matrices in the finite element literature and the vector $\mathbf{f}$ is due to the forcing terms on the right-hand side of (22). The non-linear advection term takes the form,

$$F(\mathbf{u}_i, \mathbf{b})_j = (\mathbf{u}_i)_k \mathbf{b}_l \int_D (\varphi_i)_k \frac{\partial \varphi_l}{\partial x_i} \varphi_j \, d\mathbf{x}, \quad (36)$$

which must be explicitly re-computed as the solution evolves.

To simplify the treatment of both advection and diffusion, we employ Strang splitting (Strang, 1968) to handle each separately. Specifically, we split each timestep into (1) a half-step of diffusion, (2) a full-step of advection, and (3) a final half-step of diffusion (Fig. 4). For advection, we use a second-order explicit Runge–Kutta step (also known as the midpoint method):

$$\text{Step 1: Solve (30) for } \mathbf{u}^n \text{ given } \mathbf{b}^n, \quad (37)$$

$$\text{Step 2: } \mathbf{b}_*^{n+\frac{1}{2}} = \mathbf{b}^n - \frac{\Delta t}{2} \mathbf{M}^{-1} F(\mathbf{u}_i^n, \mathbf{b}^n), \quad (38)$$

$$\text{Step 3: Solve (30) for } (\mathbf{u}_i)_*^{n+\frac{1}{2}} \text{ given } \mathbf{b}_*^{n+\frac{1}{2}}, \quad (39)$$

$$\text{Step 4: } \mathbf{b}_*^{n+1} = \mathbf{b}^n - \Delta t \mathbf{M}^{-1} F\left((\mathbf{u}_i)_*^{n+\frac{1}{2}}, \mathbf{b}_*^{n+\frac{1}{2}}\right), \quad (40)$$

where Δt is the step size. The superscripts are a short-hand for $\mathbf{b}^n = \mathbf{b}(t^n)$ where $t^n = n\Delta t$ with $n = 0, 1, 2, \dots$ and the subscript $*$ indicates that only an advection step has been performed. The matrix $\mathbf{M}^{-1}$ is not computed explicitly, but instead the linear system is iteratively solved on a GPU using the conjugate gradient (CG) method. Using a left preconditioner of $(\text{diag } \mathbf{M})^{-1}$, this approach is extremely efficient, typically converging to a reasonable tolerance in less than 10 iterations. Note that this second-order method requires two updates of the velocity field (steps 1 and 3). For the diffusion half-steps, we use the second-order accurate, semi-implicit Crank–Nicolson method:

$$\left(\mathbf{M} + \theta \frac{\Delta t}{4} \mathbf{K}\right) \mathbf{b}^{n+1} = \left(\mathbf{M} - \theta \frac{\Delta t}{4} \mathbf{K}\right) \mathbf{b}_*^{n+1} + \theta \Delta t \mathbf{f}, \quad (41)$$

We again solve this linear system using the CG method preconditioned by the inverse of the diagonal of the matrix on the left-hand side.

180 A key advantage of solving the PG equations as opposed to the full Boussinesq system is that they filter out fast-timescale dynamics, allowing for large timesteps. For the results showcased in section 4, we use a uniform timestep of $\Delta t = 0.1$, which translates to a dimensional time of about $0.1/(\Omega\varrho) \approx 100$ days. This timestep, which is orders of magnitude larger than that of most global ocean models, enables us to relatively cheaply simulate large-scale ocean dynamics over long
185 timescales. The basin simulations in the next section are run to a nondimensional time of $t = 25$ or about 80 years, with each taking only about 16 hours to complete on a single compute node with an H100 gpu. With some performance optimizations and parallelization, the vPGCM could therefore be used to investigate the millennial-timescale dynamics of the overturning circulation.

4 Results

190 With the methods laid out in the previous section, we now showcase some of the capabilities of the vPGCM. We begin by examining the effect of varying the aspect ratio α in a simple but representative example. Based on these results, we conclude that even $\alpha = 1/2$ qualitatively captures the same dynamics exhibited as $\alpha \rightarrow 0$. We then present results from a more complex simulation on a β -plane, which displays the model's ability to represent Rossby waves and complicated interior
195 buoyancy evolution. While the formulation of the model allows for a general geometry, forcings, and nondimensional parameters, we here document results for the specific case of a parabolic bowl basin with no wind stress or buoyancy forcing.

4.1 Aspect ratio effects

To examine the dynamical effect of an artificially increased aspect ratio, we run a set of simulations with $\alpha = 1/2, 1/4$, and $1/8$ in the bowl on an f -plane, i.e., $\Omega = e_3 \equiv (0, 0, 1)^T$ in nondimensional coordinates. We choose a nondimensional turbulent diffusivity profile that decays exponentially with distance (in x_3) from the bottom $z_B(\mathbf{x})$,

$$\kappa(\mathbf{x}) = 10^{-2} + \exp\left(-\frac{z_B(\mathbf{x})}{\alpha/10}\right), \quad (42)$$

qualitatively consistent with observations over rough topography (e.g., Polzin et al., 1997; Callies,
200 2018). Starting from flat isopycnals $b = x_3/\alpha$ and no flow, a weakly stratified layer near the bottom develops to satisfy the no-flux boundary condition, bending isopycnals into the slopes (gray lines in Fig. 5). The cross-slope buoyancy gradient within this layer generates strong upwelling in the bottom BL and weak downwelling aloft, characteristic of bottom-enhanced mixing (Fig. 5a,c). Due to the rotational symmetry of this geometry on an f -plane, the net transport of this secondary cross-slope

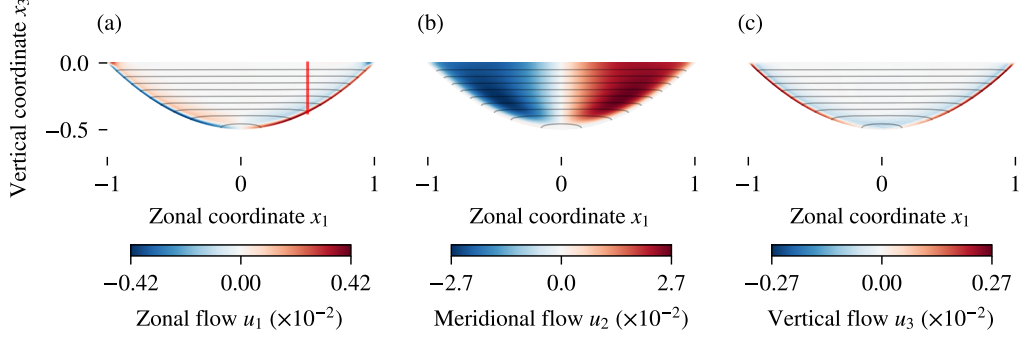


Figure 5. Snapshot of flow components (colors) and isopycnals (gray lines) at $x_2 = 0$ and nondimensional time $t = 25$ in a three-dimensional simulation of mixing-driven spin up on an f -plane ($\Omega = e_3$). The aspect ratio is $\alpha = 1/2$, Ekman number is $\epsilon = 2 \times 10^{-2}$, and Prandtl times Burger number is $\mu\varrho = 1$. The vertical red line in (a) indicates location of profiles in Fig. 1.

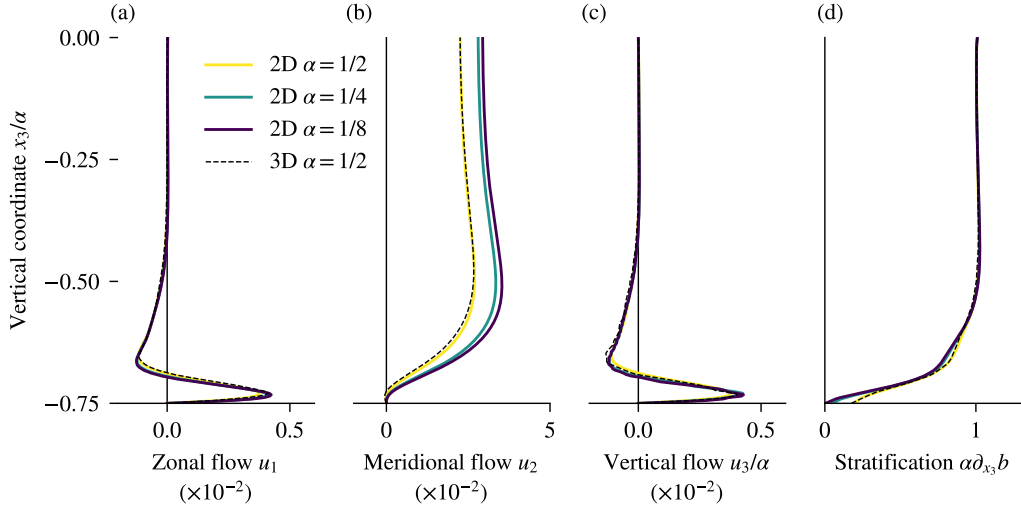


Figure 6. Vertical profiles taken at $x_1 = 0.5$ and $x_2 = 0$ in the bowl (red line in Fig. 5) at nondimensional time $t = 25$ for spin up on an f -plane ($\Omega = e_3$). The panels show the (a) zonal flow u_1 , (b) meridional flow u_2 , (c) vertical flow u_3/α , and (d) stratification $\alpha \partial_{x_3} b$, with solid lines for two-dimensional simulations and dashed lines for three-dimensional. Results are shown for an aspect ratio α of $1/2$, $1/4$, and $1/8$, with the vertical coordinate scaled by $1/\alpha$ to allow for a direct comparison.

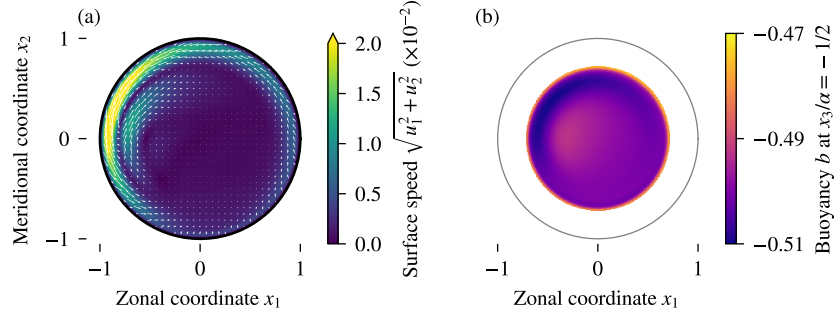


Figure 7. Snapshots at nondimensional time $t = 25$ of (a) velocity (arrows) and speed (colors) at the surface ($x_3 = 0$) and (b) buoyancy at $x_3/\alpha = 1/2$ in a three-dimensional simulation of mixing-driven spin up on an β -plane [$\Omega = (1 + x_2)e_3$]. The aspect ratio is $\alpha = 1/2$, Ekman number is $\epsilon = 2 \times 10^{-2}$, and Prandtl times Burger number is $\mu\varrho = 1$. Gray circle in (b) represents the coastline at $x_3 = 0$.

circulation must be zero. By thermal wind balance, the horizontal buoyancy gradients also support substantial shear in the meridional flow near the bottom, leading to a strong circumbasin flow in the interior (Fig. 5b). This flow has little effect on the buoyancy evolution since along-slope buoyancy gradients are weak. The cross-slope BL upwelling, on the other hand, brings dense water up the slope, working against mixing to bolster the stratification near the bottom. Without any buoyancy forcing, however, the simulation tends towards a completely mixed state.

Because this geometry is axisymmetric on an f -plane, the dynamics are well-described by a two-dimensional model in polar coordinates with no variations in the azimuthal direction (Peterson & Callies, 2022, 2023). Even for a two-dimensional simulation in Cartesian coordinates, the flow and stratification are nearly identical to the three-dimensional bowl (compare solid to dashed lines in Fig. 6), providing a useful validation of the model. As the aspect ratio is decreased, some quantitative changes in the solution emerge, with the largest deviations occurring in the far-field along-slope flow (Fig. 6b) and the stratification near the bottom (Fig. 6d). The qualitative structure of the circulation, however, remains robust even for $\alpha = 1/2$. For the purposes of most idealized experiments, we therefore do not expect the artificially increased aspect ratio to affect qualitative conclusions. A more quantitative study of the overturning with realistic bathymetry and forcing, however, would warrant using an aspect ratio as small as computationally feasible.

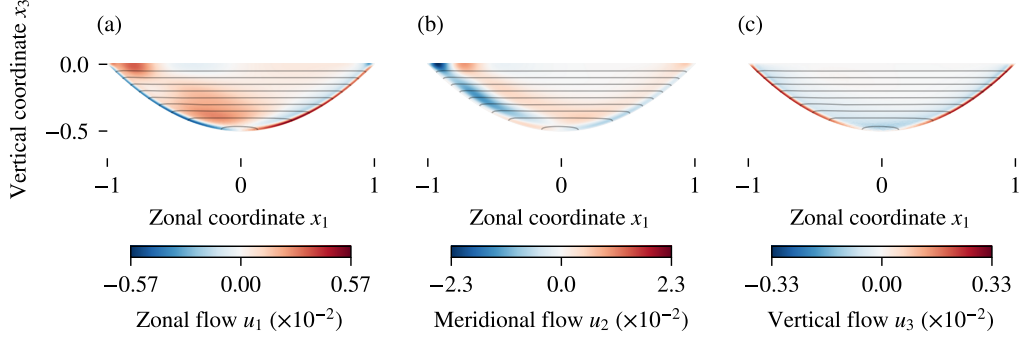


Figure 8. Snapshot of flow components (colors) and isopycnals (gray lines) at $x_2 = 0$ and nondimensional time $t = 25$ in a three-dimensional simulation of mixing-driven spin up on an β -plane [$\Omega = (1 + x_2)e_3$]. The aspect ratio is $\alpha = 1/2$, Ekman number is $\varepsilon = 2 \times 10^{-2}$, and Prandtl times Burger number is $\mu\varrho = 1$.

4.2 Circulation in a bowl basin on a β -plane

For a more dynamic showcase of the capabilities of the model, we solve for the dynamics on a β -plane, varying the rotation rate in the meridional direction such that $\Omega = (1 + x_2)e_3$. This breaks the along-slope symmetry that was present in the f -plane setup, leading to a circulation that can no longer be described by two-dimensional physics. One particularly striking difference between the two simulations is the presence of Rossby waves on the β -plane (Figs. 7 and 8). The early evolution resembles the f -plane case, with mixing bending isopycnals into the slope, generating zonal buoyancy gradients that support a shear in the meridional flow by thermal wind balance. This initial perturbation in the meridional flow then excites Rossby waves that propagate westward along the planetary vorticity gradient. These waves eventually reach the western side of the basin, where they interact with the boundary and set up a western boundary current (e.g., Pedlosky et al., 1997). Other PG ocean models also set up western boundary currents (e.g., Callies & Ferrari, 2018), but Rayleigh drag tends to suppress Rossby waves more aggressively than the Fickian friction used here (Callies, personal communication).

5 Conclusions and Future Directions

The vPGCM represents a novel use of finite elements in a numerical model of the large-scale ocean circulation, allowing it to capture the complicated geometry of the ocean and accurately represent the near-bottom dynamics that are now understood to be a crucial component of the overturning. By making the PG approximation, which filters out fast, small-scale dynamics, the model can afford

to take large timesteps needed to simulate the overturning on millennial timescales. This unique tool could therefore help answer many outstanding questions about the large-scale ocean circulation. For instance, this model could be used to explain how the overturning may have shifted during the Last Glacial Maximum to store more carbon (Curry & Oppo, 2005; Sigman et al., 2010; Lund et al., 2011; Ferrari et al., 2014; Jansen, 2017). It could also be used to uncover the mechanisms behind and implications of the observed abyssal heat uptake in recent decades (Purkey & Johnson, 2010; Desbruyères et al., 2016; Lele et al., 2021; Johnson & Purkey, 2024). Ultimately, the ν PGCM could even be coupled to ice, atmosphere, and land models to understand the full Earth system (e.g., Holden et al., 2016).

We also envision the numerical approaches developed here as potentially paving the way for the more general adoption of finite elements and unstructured grids in the field. The methods described in section 3 could naturally be extended to the full Boussinesq system by reformulating the inversion as an Oseen problem iteration. The aspect ratio trick could then be the key to leveraging standard mixed finite element methods in global ocean models without the need for complicated stabilization schemes.

6 Open Research

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